BASIC CLOSURE PROPERTIES OF THE COVERING SIEVES OF A TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The axioms of a topology. Same chapter, Definition definition-topology. Supplies the axioms used in both verifications.
- The pullback of a sieve along a morphism into a member. Same chapter, Lemma lemma-pullback-sieve-section. Identifies the pullback of a sieve along a morphism that already belongs to it.

Source reference. Original definition 00Z4: sources/stacks-sites/pinned-original-source.tex, source lines 11244–11244. Navigation pointer invoked at proof line 11244; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00Z3: sources/stacks-sites/pinned-original-source.tex, source lines 11256–11256. Navigation pointer invoked at proof line 11256; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let J be a topology on $\mathcal{C}$. Let $U \in \mathrm{Ob}(\mathcal{C})$.*

(1) *Finite intersections of elements of $J(U)$ are in $J(U)$.*
(2) *If $S \in J(U)$ and $S' \supset S$, then $S' \in J(U)$.*

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Proof. Proof of (1). The intersection of the empty family follows from condition (3) of Definition 00Z4. Let $S, S' \in J(U)$. Consider $S'' = S \cap S'$. For every $V \to U$ in $S(V)$ we have

$$S' \times_U V = S'' \times_U V$$

simply because $V \to U$ already is in S. Hence by the second axiom of the definition we see that $S'' \in J(U)$.

Proof of (2). Let $S \in J(U)$ and $S' \supset S$. For every $V \to U$ in $S(V)$ we have $S' \times_U V = h_V$ by Lemma 00Z3. Thus $S' \times_U V \in J(V)$ by the third axiom. Hence $S' \in J(U)$ by the second axiom. $\square$